



# SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY

## COMPUTER ENGINEERING DEPARTMENT

### Proposal for the Final Year Project Batch 2022f



| Group Members Name | Roll No       | Section | GL Name | Contact No: |
|--------------------|---------------|---------|---------|-------------|
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### 1. Project Title:

**LogAI – AI-Powered Log Monitoring Dashboard**

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### 2. Brief Project Description:

**LogAI** is an intelligent, cloud-based monitoring dashboard that leverages **Artificial Intelligence (AI)** to analyze system and application logs from diverse sources — including **Kubernetes clusters, Docker containers, PM2 Node.js apps, and traditional servers**. The system automates log analysis, detects unusual behavior, predicts potential system failures, and provides **real-time alerts with recommended fixes**.

The platform is composed of four primary modules:

- DevOps Module:** Handles infrastructure setup, deployment automation, and log ingestion via Fluentd/FluentBit agents.
- Backend Module:** Provides REST APIs for log storage, alerting, and analytics using Node.js and Django.
- Frontend Dashboard:** Developed in React.js and Tailwind CSS to display real-time logs, anomaly charts, and AI alerts.
- AI/ML Module:** Implements models such as **Isolation Forest** and **LSTM** for anomaly detection and predictive monitoring.

This project delivers a **scalable, enterprise-grade monitoring tool** that empowers DevOps teams, developers, and system administrators to visualize, detect, and mitigate failures across distributed systems — improving reliability and reducing downtime.

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### 3. Project Category:

☒ **Capstone Project — Integrates AI, DevOps, and Full Stack Development**

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#### 4. Complex Engineering Problem (CEP) Characteristics Applicable:

- ✓ **Range of conflicting requirements:** balancing scalability, large data volume, AI model accuracy, and low latency.
  - ✓ **Depth of analysis required:** advanced understanding of data pipelines, anomaly detection, and cloud deployment.
  - ✓ **Depth of knowledge required:** integrates AI/ML models, DevOps automation, and full stack visualization.
  - ✓ **Familiarity of issues:** real-time AI-based monitoring is rarely implemented in academic environments.
  - ✓ **Extent of applicable codes:** involves containerization standards, security protocols, and cloud monitoring APIs.
  - ✓ **Consequences:** ensures high system reliability, reduces production downtime, and enhances application stability.
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#### 5. Brief Objectives/Functionalities Completed Before Mid Evaluation:

- Set up **Kubernetes cluster** and configure **CI/CD pipeline** for automated deployment.
  - Develop **Node.js API** for receiving and processing logs from multiple agents.
  - Integrate **Elasticsearch/OpenSearch** for efficient log indexing and querying.
  - Build **React.js dashboard** for displaying live logs and basic statistics.
  - Implement **WebSocket-based real-time log streaming** between backend and frontend.
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#### 6. Brief Objectives/Functionalities Completed After Mid Evaluation:

- Implement **AI-based anomaly detection** for automated pattern recognition.
  - Integrate **alert notification system** via Slack and Email.
  - Enhance dashboard with **real-time charts, search, and filtering capabilities**.
  - Deploy entire system on **Kubernetes** with monitoring and autoscaling support.
  - Perform **performance testing, documentation, and final project presentation**.
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#### 7. Scope:

**LogAI** aims to simplify large-scale monitoring through AI-assisted automation. It provides a centralized platform for visualizing logs, identifying anomalies, and predicting failures in complex environments.

The project primarily targets **DevOps teams, software houses, and cloud service providers** that rely on continuous uptime. In future iterations, **LogAI** can be expanded to include **distributed tracing, multi-cloud observability, and integration with external tools** such as Grafana, Prometheus, and Datadog.

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#### 8. Methodology:

1. **Research & Planning:** Review existing monitoring tools, log formats, and AI anomaly detection algorithms.



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- Design Phase:** Create modular system architecture integrating backend APIs, AI engine, and frontend dashboard.
- Development Phase I:** Implement log ingestion pipeline (Fluentd → Node.js → Elasticsearch).
- Development Phase II:** Train ML models for anomaly detection and integrate them with backend APIs.
- Frontend Integration:** Build real-time React.js dashboard using WebSocket streaming.
- Testing & Validation:** Conduct scalability, latency, and model accuracy testing.
- Deployment:** Deploy complete system on Kubernetes using Docker, GitHub Actions, and ArgoCD.

## 9. Expected Timeline (10–12 Months):

| Phase      | Duration                | Activities                                                                          |
|------------|-------------------------|-------------------------------------------------------------------------------------|
| Month 1–2  | Research & Design       | Study DevOps architecture, logging agents, and AI techniques.                       |
| Month 3–4  | Infrastructure Setup    | Configure Kubernetes cluster, Elasticsearch, and CI/CD pipeline.                    |
| Month 5–6  | Backend Development     | Develop Node.js/Django APIs, database schema, and log ingestion.                    |
| Month 7–8  | Frontend Integration    | Build React dashboard with WebSocket streaming and visual charts.                   |
| Month 9–10 | AI Integration          | Train anomaly detection models (Isolation Forest, LSTM) and integrate with backend. |
| Month 11   | Testing & Optimization  | Conduct stress testing, scalability validation, and optimize performance.           |
| Month 12   | Deployment & Evaluation | Deploy full system, finalize documentation, and prepare final presentation.         |

## 10. Expected Budget:

| Item                              | Estimated Cost (PKR)          |
|-----------------------------------|-------------------------------|
| Cloud Hosting / VM Instances      | 6,000                         |
| Elasticsearch & Database Hosting  | 4,000                         |
| AI Model Training (GPU/Colab Pro) | 3,000                         |
| Miscellaneous Tools / APIs        | 2,000                         |
| <b>Total Estimated Budget</b>     | <b>≈ 15,000 PKR (~USD 50)</b> |

## 11. Project Output:

- Fully functional **AI-powered log monitoring dashboard**.
- Real-time anomaly detection** with predictive alerts.
- Automated CI/CD** deployment on Kubernetes.



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- Integrated **backend APIs** with live WebSocket data streaming.
- Insightful **charts and metrics** for performance monitoring and diagnostics.

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## 12. Hardware/Components Used:

- Cloud-based Virtual Machines (AWS/GCP).
- Developer PCs for local testing.
- Docker containers for isolated development environments.
- Kubernetes cluster for system deployment and autoscaling.

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## 13. Software Tools Used:

- **Frontend:** React.js, Tailwind CSS, Chart.js, WebSocket
- **Backend:** Node.js (Express), Django REST Framework, PostgreSQL/Firebase/MySQL
- **AI/ML:** Python, Scikit-learn, Pandas, Isolation Forest, LSTM
- **DevOps:** Docker, Kubernetes, GitHub Actions, Linux,
- **Database/Logs:** Elasticsearch or OpenSearch

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## 14. Alignment with Departmental Mission and SDGs:

### Departmental Alignment:

The project aligns with the department's mission to develop engineers proficient in **AI-driven software systems**, **DevOps automation**, and **scalable backend architectures**. It combines multiple computing disciplines — AI, cloud computing, and software engineering — into a cohesive enterprise solution.

### Relevant SDGs:

- **SDG 9 – Industry, Innovation & Infrastructure:** Builds resilient, intelligent infrastructure for organizations.
- **SDG 8 – Decent Work & Economic Growth:** Enhances system reliability, reducing operational disruptions in tech ecosystems.
- **SDG 12 – Responsible Consumption & Production:** Encourages efficient resource monitoring and sustainable automation.



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### Proposal for the Final Year Project Batch 2022f

Name of Proposed Supervisor:

Signature:

Name of Proposed Co- Supervisor (if any):

Signature: \_\_\_\_\_

Name of Industrial Advisor/External (if any):

**Recommendations and Approval:**

**Chairperson Sign with Date:**

**Head of FYDP Committee Sign with Date:**